

Lab 2 | Active Directory Domain Setup & Windows 11 Domain Join

Portfolio focus: Windows Server administration | Active Directory Domain Services | Domain integration | IAM foundations

Project overview

Built a Windows Server 2022 Active Directory lab environment using VMware and worked through joining a Windows 11 client to the domain. This environment provided the foundation for later identity, group access, and authentication labs.

Lab environment

Component	Lab configuration
Hypervisor	VMware Workstation Player 17
Domain controller	JMC-DC01 Windows Server 2022
Active Directory domain	jmc-local
Client	Windows 11 virtual machine
Networking	VMware NAT (VMnet8); private lab network

Objectives

- Establish a Windows Server domain controller and Active Directory domain.
- Configure virtual network connectivity for domain services.
- Join a Windows 11 client to the Active Directory domain.
- Establish a reusable foundation for user provisioning, group management, and access-control labs.

Workflow documented

- 1. Prepare the environment** — Configure Windows Server 2022 and Windows 11 virtual machines in VMware with a shared lab network.
- 2. Configure directory services** — Set up the domain controller and Active Directory domain jmc-local.
- 3. Validate network and DNS** — Check that the Windows 11 client can resolve and reach the domain controller using the lab network.
- 4. Domain integration** — Work through joining the Windows 11 client to the jmc-local domain and using domain credentials.
- 5. Verify readiness** — Confirm that the environment is suitable for later Active Directory and IAM exercises.

Security concepts demonstrated

- Centralized identity and authentication with Active Directory Domain Services.

- Domain-based administration and client trust relationships.
- DNS and network connectivity as dependencies for authentication.
- Separation of lab infrastructure from production environments.

Outcome and evidence

The domain controller and domain were used in subsequent IAM labs involving Active Directory users, security groups, and shared-folder access. Original Lab 2 setup and domain-join screenshots have not been included in this version; the document does not independently verify every configuration step.

Suggested evidence to add later: Server computer name/domain properties; Active Directory Users and Computers; Windows 11 domain membership; a successful domain-user sign-in. Avoid publishing passwords or sensitive identifiers.

Portfolio note

This is a concise project summary based on the recorded lab environment. It is not a substitute for configuration screenshots or command output.